

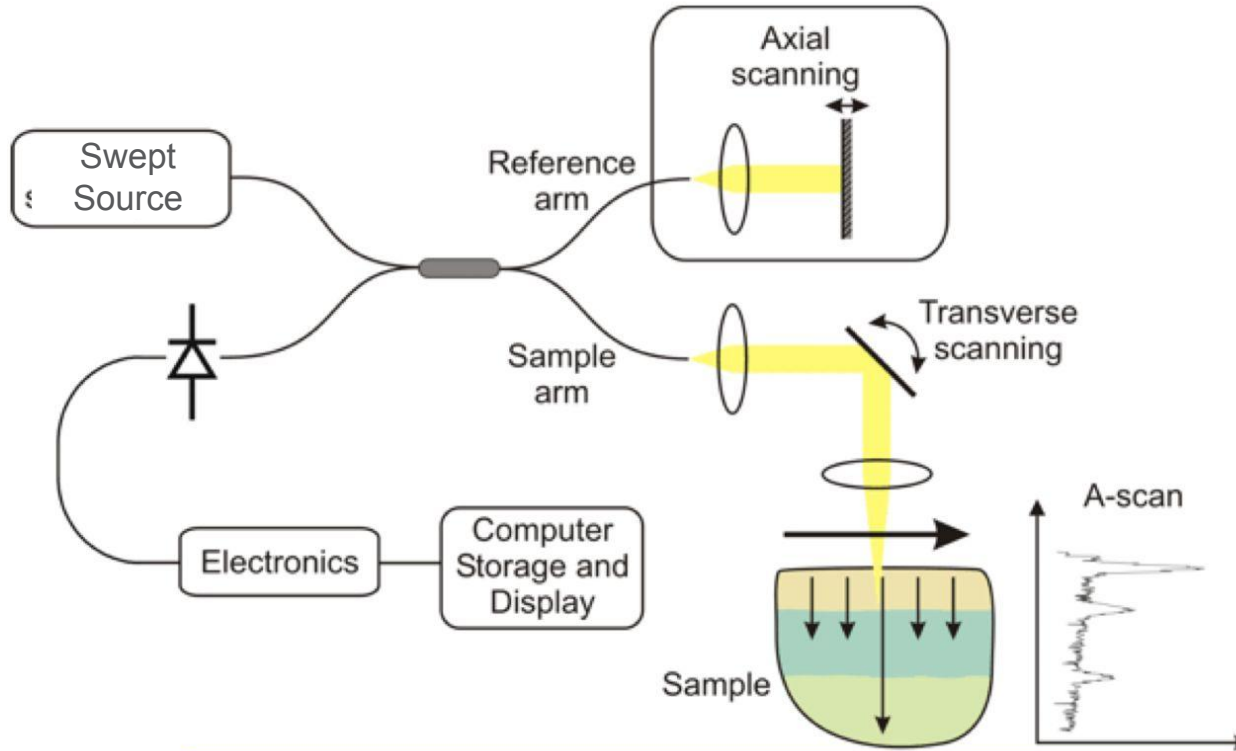


Samueli
School of Engineering

Revolutionizing Optical Coherence Tomography with AI

Zaid Slayyeh
Bioengineering

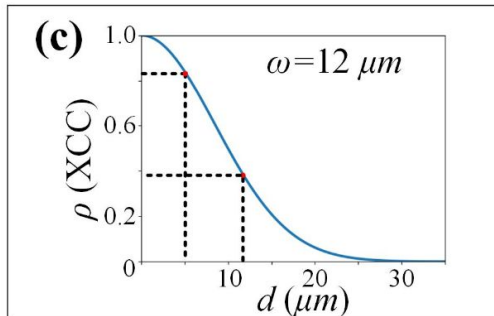
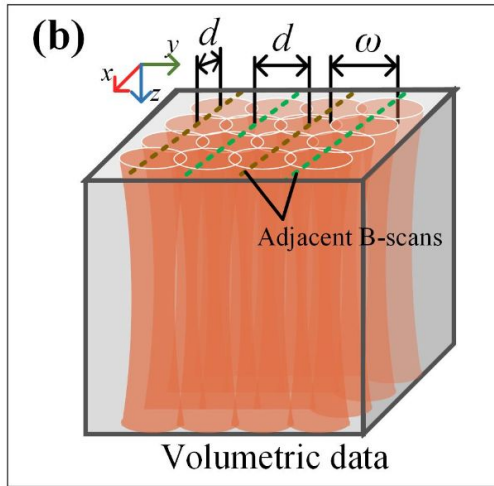
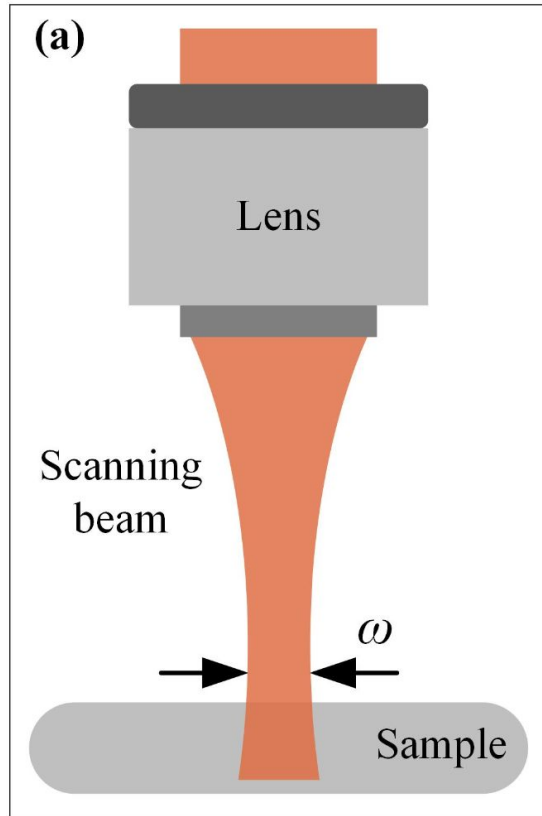
Quick Optical Coherence Tomography Refresher



Back Scattered Waves Analyzed
→ *Gives Depth*

Interfering Light Measured
→ *Reveals Structure*

Making a Volumetric Image



$$\rho = \exp\left(-\frac{d^2}{\omega^2}\right)$$

ω = Gaussian Beam Waist

d = Lateral Distance

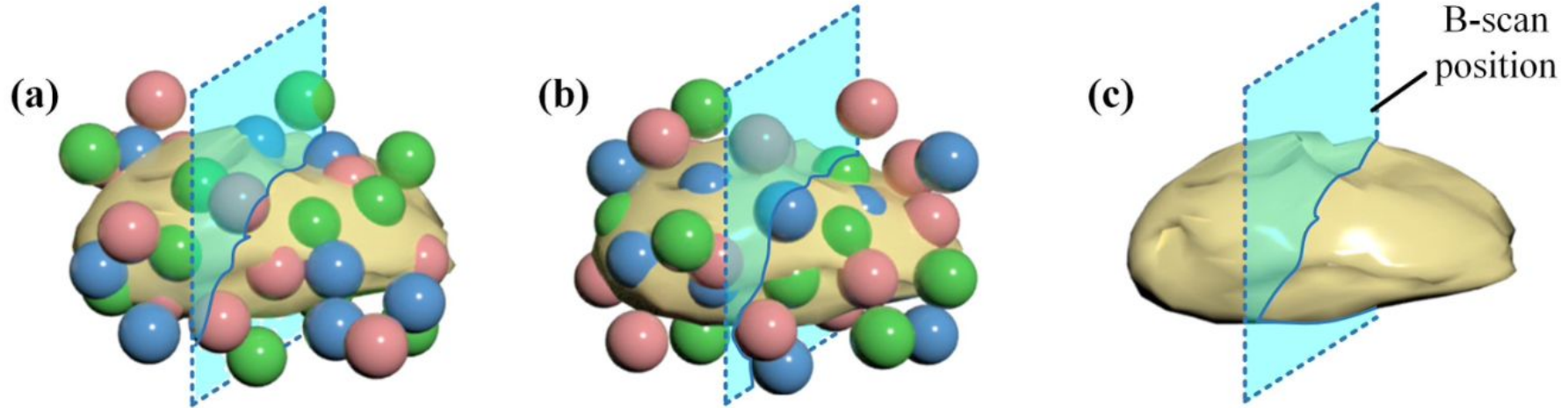
XXC = Cross Correlation Coefficient

ρ = Correlation between two A-Scans

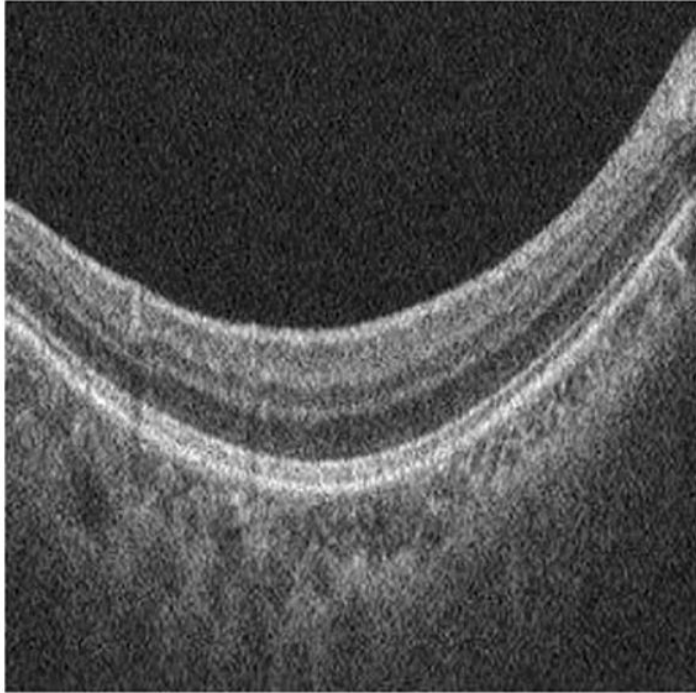
Speckle is a Big Problem in OCT

Speckle Noise is Unavoidable

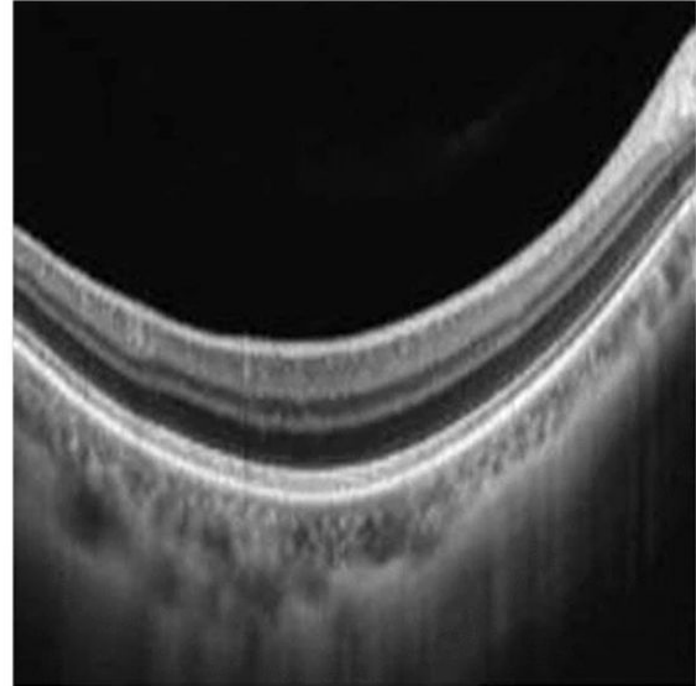
Speckle noise → Random light scattering and interference



What Speckle Looks like in OCT



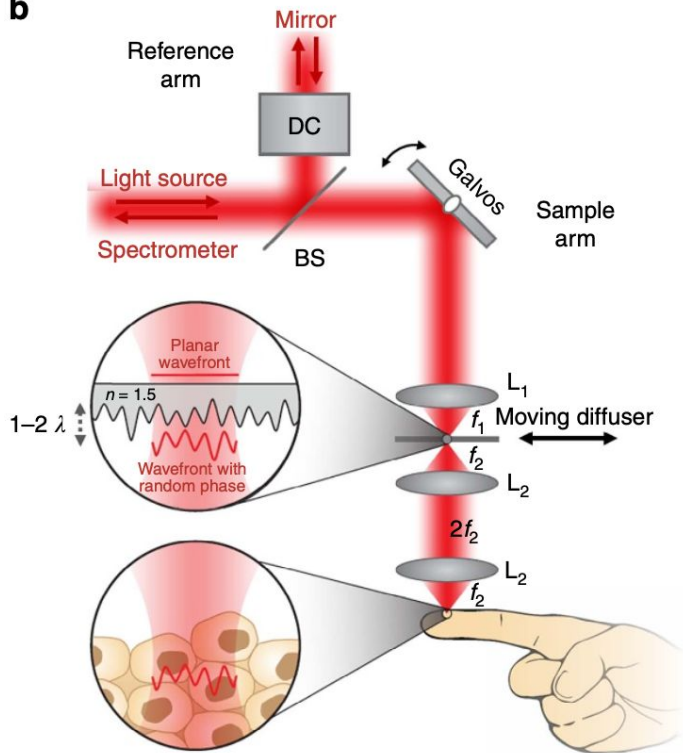
(a) Original



(b) Ground truth

SM-OCT as the SOTA Denoising Method

b



The Gold Standard

Light Manipulation:

→ Moving Diffuser Introduces Phase Shifts

Averages Scans of Uncorrelated Speckle:

→ Eliminates Noise (*Near All*)

Needs to Perform Repeated B-scans at Same Location:

→ Unsuitable for Real-Time imaging

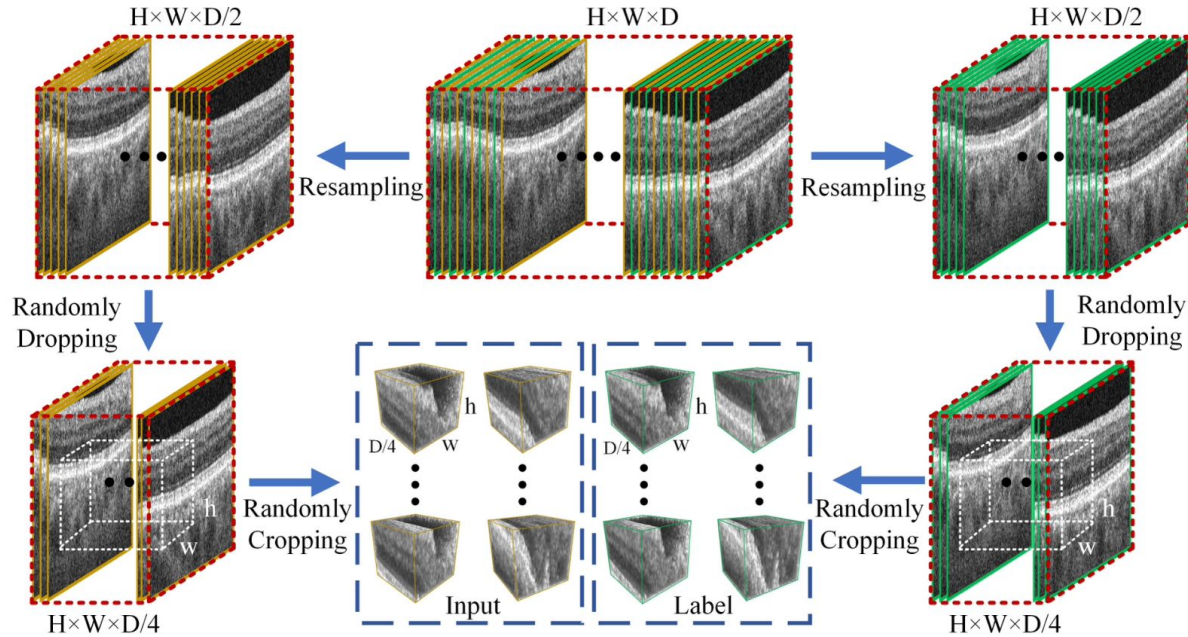
→ High/Costly Data Burden

Diving into the Paper

“Toward ground-truth optical coherence tomography via three-dimensional unsupervised deep learning processing and data”

- ➡ Using a GAN to optimize imaging
- ➡ tGT-OCT to reduce speckle noise and reveal obscured/undetectable structures
- ➡ Preserve spatial resolution
- ➡ No Ground Truth Needed

Noise2Noise Method Breakdown



Used for Training and Testing

→ 3D Data put into Two Sub-Datasets

- Odd Set (1st, 3rd, 5th...)
- Even Set (2nd, 4th, 6th...)

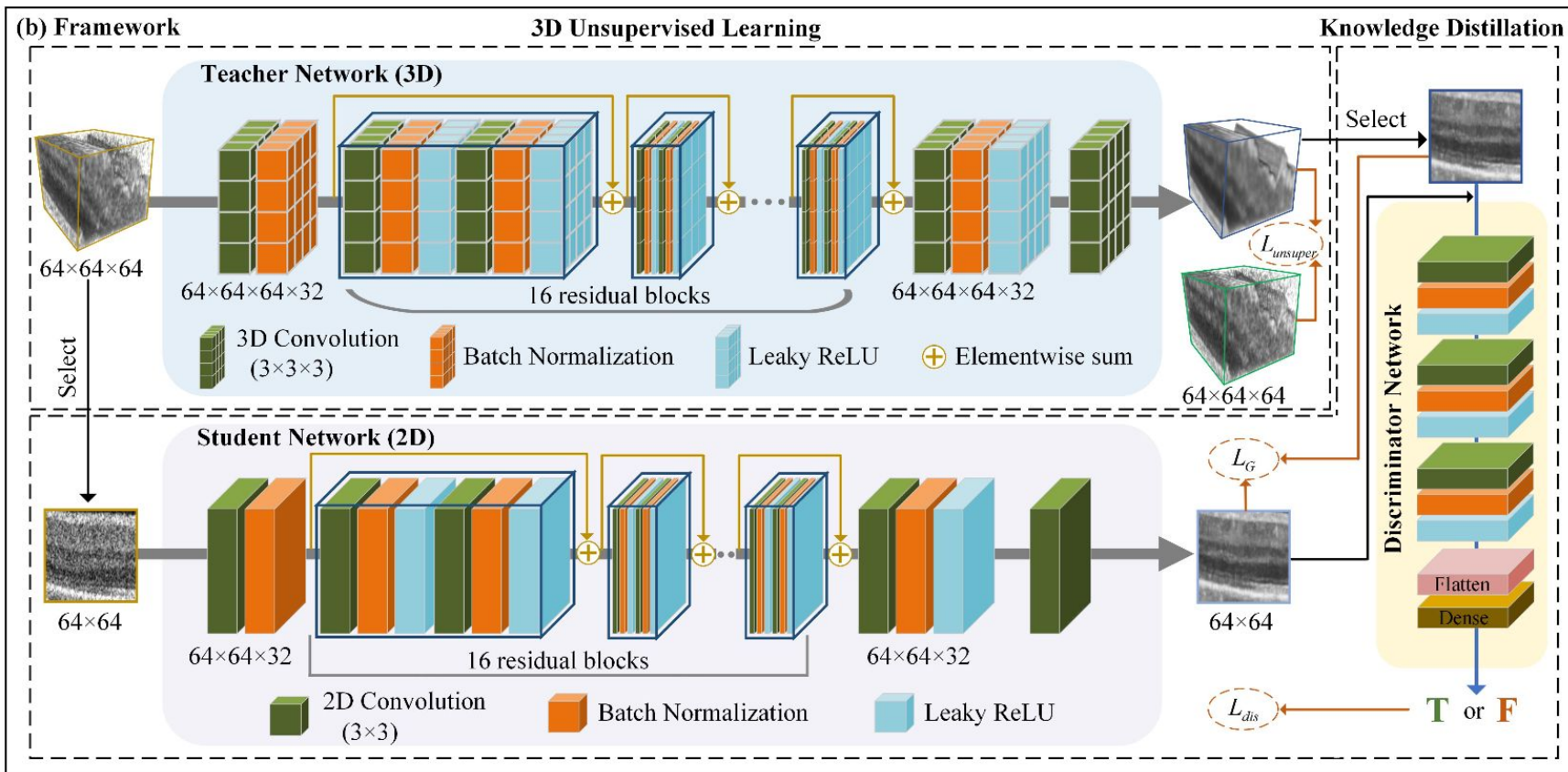
→ Random B-Scan Dropping

- One of Two Scans
- Two of Four Scans

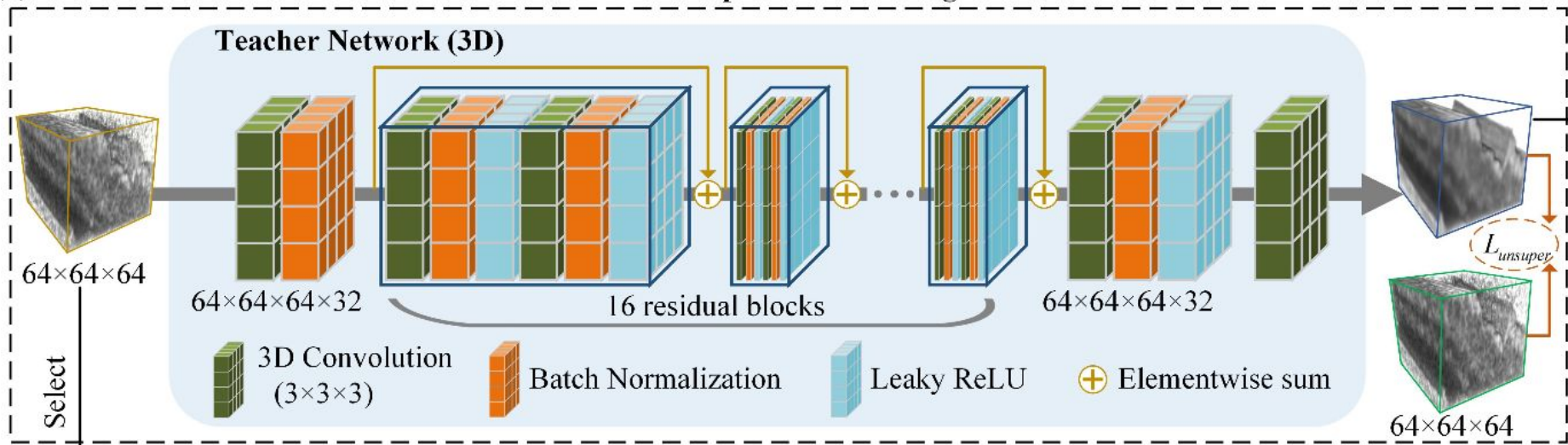
→ Random Cropping

- Cropping Block Size ($h, w, D/4$)

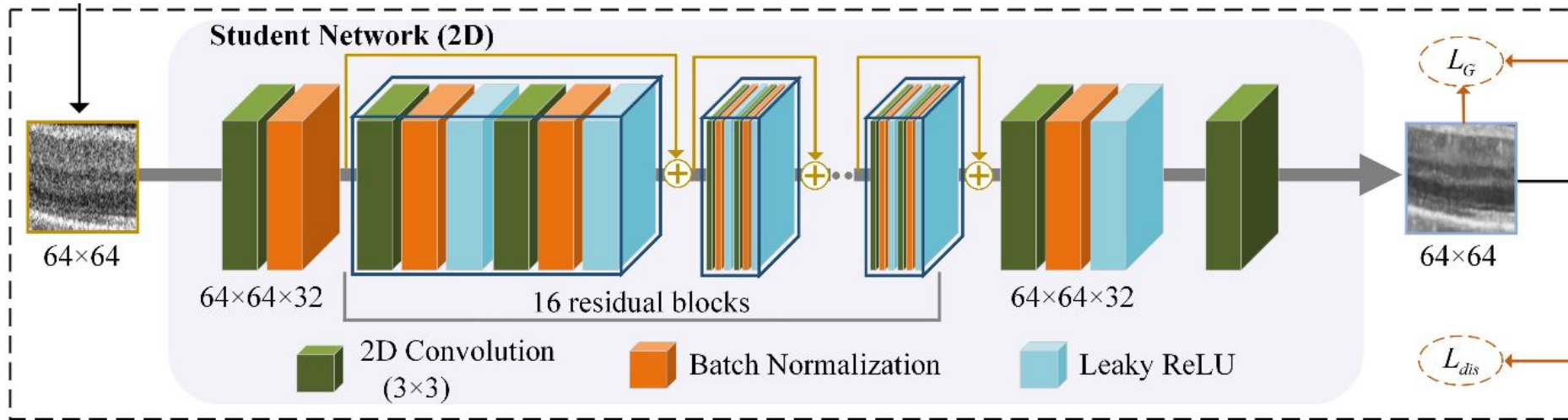
Overview of the Framework



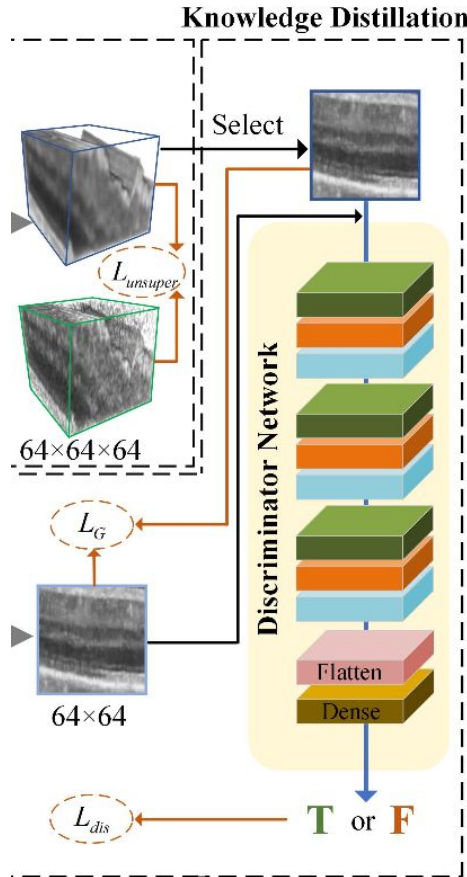
Deeper into the 3D Teacher Network



Deeper into the 2D Student Network



Deeper into the Discriminator Network



Loss Functions

The unsupervised learning process of the 3D CNN:

$$L_{unsuper} = \frac{1}{HWF} \sum_{i,j,k=1}^{H,W,F} \|v_{i,j,k} - p_{i,j,k}\|^2$$

A Noise2Noise Method

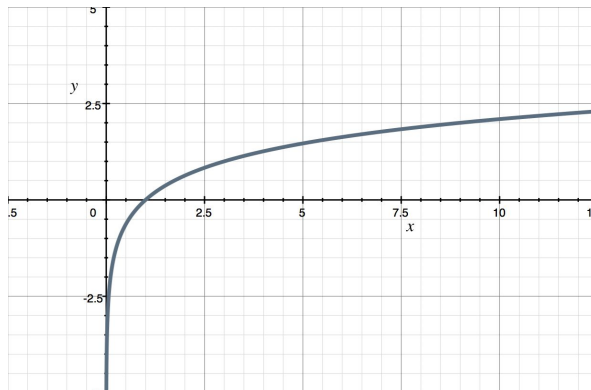
Optimizing the parameters of the 3D CNN during training

$v_{i,j,k}$ and $p_{i,j,k}$ = are pixels of two generated similar noisy OCT volumetric data

Loss Functions - Generative Adversarial Network

$$\min_G \text{Loss}_{adv}(G) = \frac{1}{HW} \sum_{i,j=1}^{H,W} \log(1 - D(G(x_{i,j})))$$

$x_{i,j}$ = Pixels of the Generated Image (From the 2D Network)



G = Generator Network

Loss Functions - Generative Adversarial Network

$$\min_D \text{Loss}_{dis}(D) = \frac{1}{HW} \sum_{i,j=1}^{H,W} \log(D(y_{i,j})) + \frac{1}{HW} \sum_{i,j=1}^{H,W} \log(D(G(x_{i,j})) - 1)$$

$y_{i,j}$ = Pixels of the Ground Truth (From the 3D Network)

G = Generator Network

Loss Functions - Generative Adversarial Network

$$\min_D \text{Loss}_{dis}(D) = \frac{1}{HW} \sum_{i,j=1}^{H,W} \log(D(x_{i,j})) + \frac{1}{HW} \sum_{i,j=1}^{H,W} \log(D(G(x_{i,j}))) - 1$$

$y_{i,j}$ = Pixels of the Ground Truth (From the 3D Network)

G = Generator Network

Loss Functions - Generative Adversarial Network

~~$$\min_D \text{Loss}_{dis}(D) = \frac{1}{HW} \sum_{i,j=1}^{H,W} \log(D(y_{i,j})) + \frac{1}{HW} \sum_{i,j=1}^{H,W} \log(D(G(x_{i,j})) - 1)$$~~

$$\min_D \text{Loss}_{dis}(D) = - \left\{ \frac{1}{HW} \sum_{i,j=1}^{H,W} \log(D(y_{i,j})) + \frac{1}{HW} \sum_{i,j=1}^{H,W} \log(1 - D(G(x_{i,j}))) \right\}$$

$y_{i,j}$ = Pixels of the Ground Truth (From the 3D Network)

G = Generator Network

Comparing tGT-OCT with SOTA

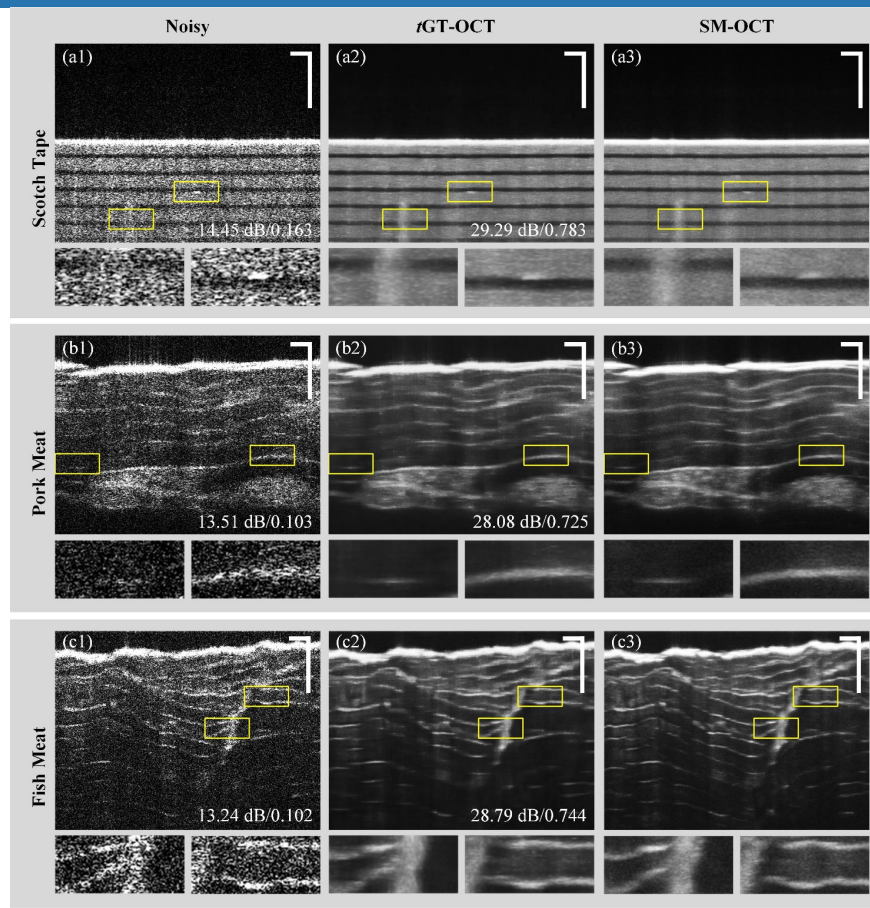
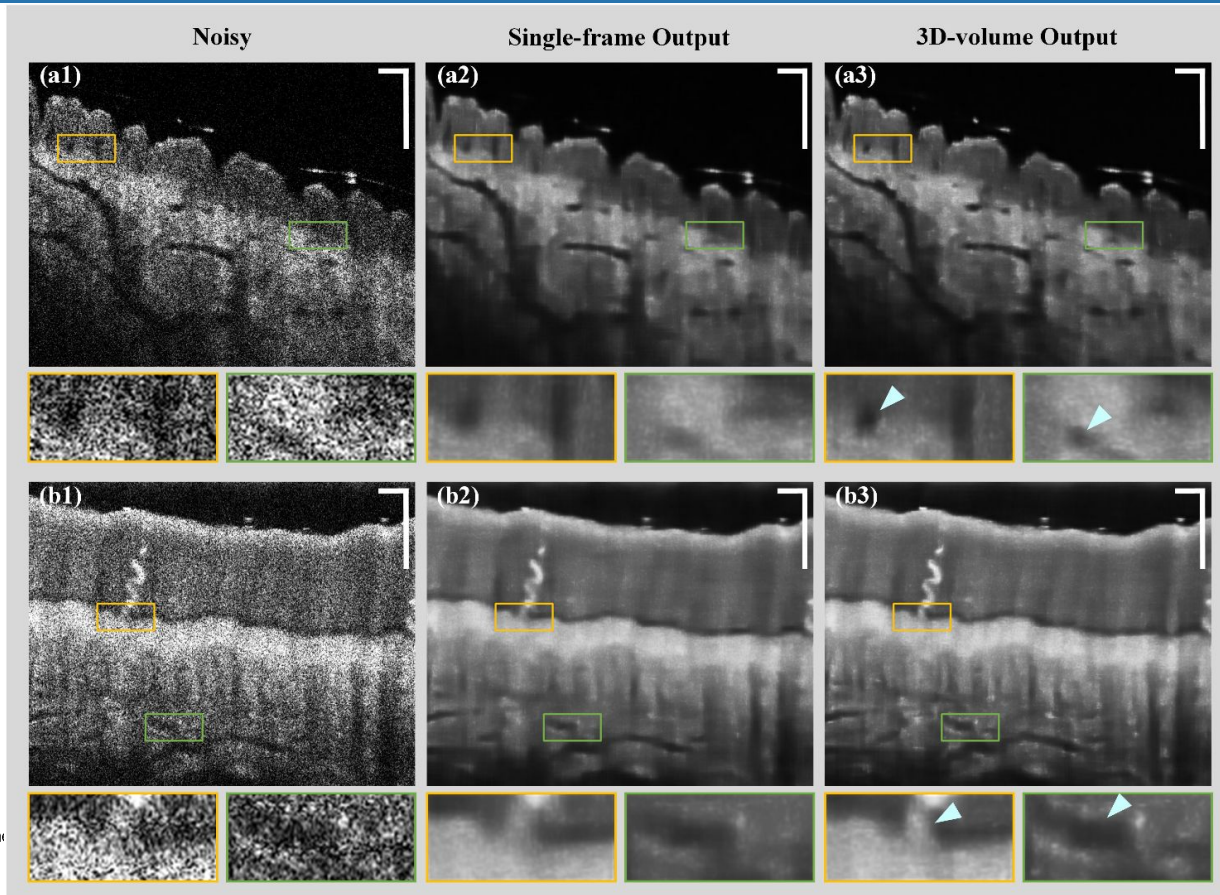


Image outputs of the proposed tGT-OCT method, and comparison with SM-OCT images of Scotch tape, pork and fish.

The evaluation metrics format is PSNR/SSIM.

Scale bar: 200 μ m.

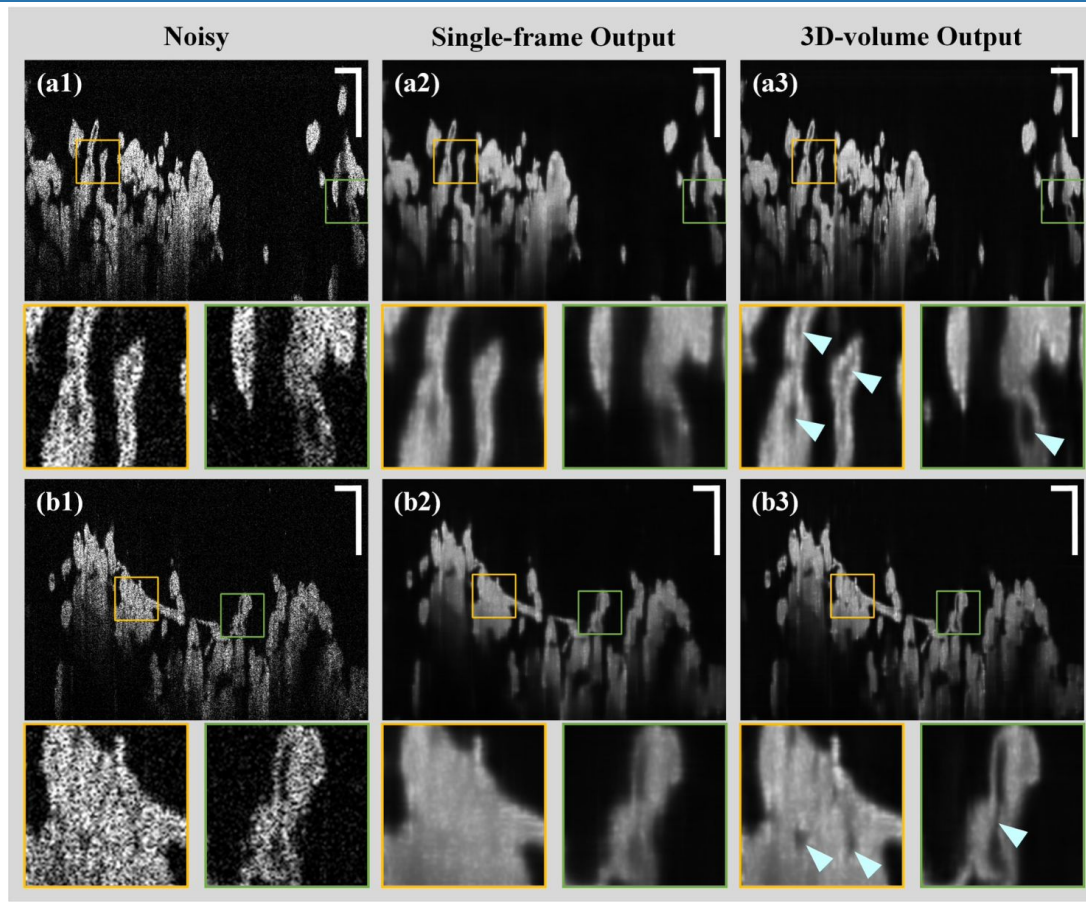
Comparing the 3D and 2D Network Outputs



Denoised images of the **human skin** output by the proposed tGT-OCT.

Scale bar: 200 μm .

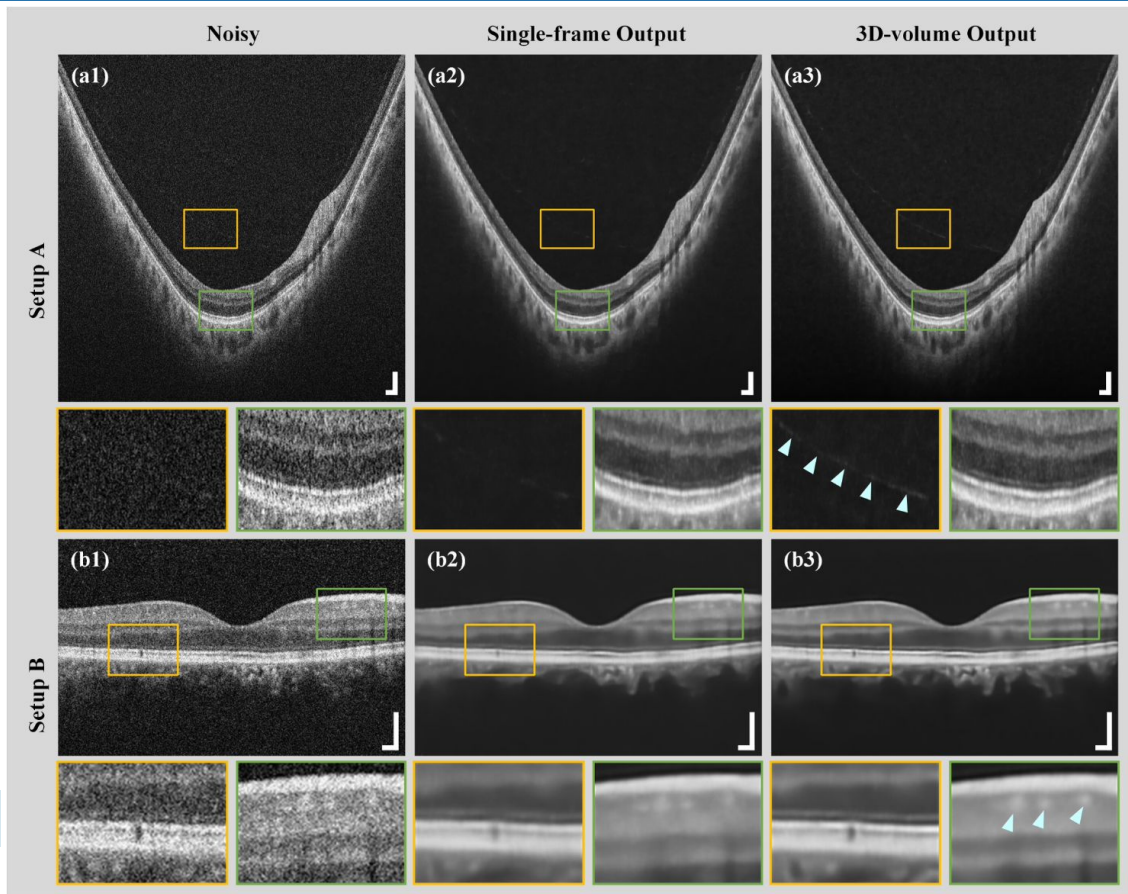
Comparing the 3D and 2D Network Outputs



Denoised images of the **human placenta** output by the proposed tGT-OCT.

Scale bar: 200 μm .

Testing Generalization of tGT-OCT



(Setup A)

(a1) is an image from the **OCT-R1** Dataset with a **BM-400K BMizar** scanner

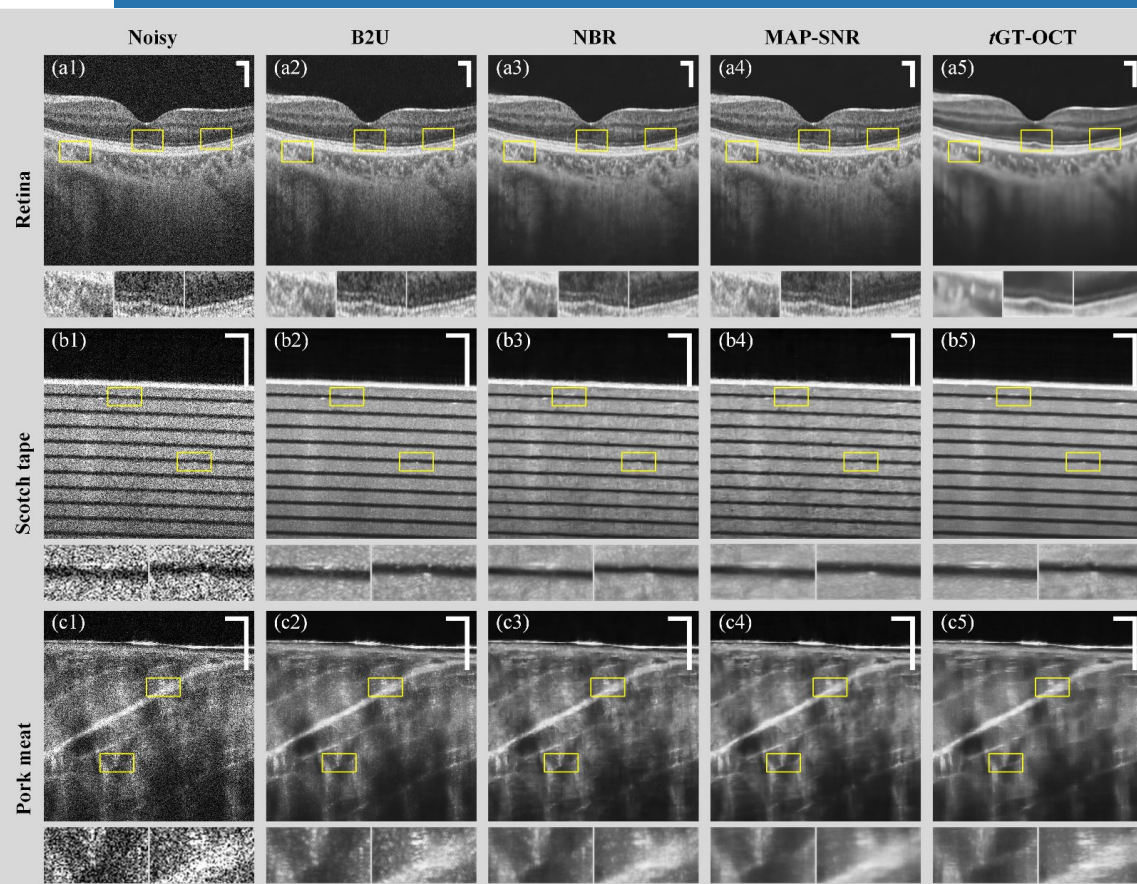
(Setup B)

(b1) is an image of the **OCT-R2** dataset obtained with a **Spectralis OCT** system

Network had never seen images from the OCT-R2 dataset during training

Scale bar: 200 μ m.

Comparing Against Other Deep Learning Methods



B2U = Blind2Unblind

NBR = Neighbor2Neighbor

MAP-SNR = Mapping between
Adjacent Pixels and eventually work
for Speckle Noise Reduction

Comparison of the denoising performance
of different deep-learning-based methods
with that of tGT-OCT.

Scale bar: 200 μ m.

Quantitative Comparisons

		Noisy	B2U	NBR	MAP-SNR	tGT-OCT
Human Retina	SNR	12.183	30.630	42.391	45.093	<u>48.214</u>
	CNR	1.628	4.085	4.756	4.602	<u>5.817</u>
	ENL	1.064	61.063	1032.887	1964.520	<u>3258.357</u>
Scotch Tape	SNR	17.795	39.490	42.159	40.678	<u>42.994</u>
	CNR	2.463	6.248	6.673	6.624	<u>6.718</u>
	ENL	0.393	57.594	117.019	69.625	<u>139.210</u>
Pork Meat	SNR	16.206	42.247	41.189	39.081	<u>43.895</u>
	CNR	1.837	5.694	6.070	6.261	<u>6.568</u>
	ENL	0.342	128.632	115.372	59.551	<u>193.111</u>

Quantitative comparison of tGT-OCT and different deep-learning denoising methods on different samples

B2U = Blind2Unblind

NBR = Neighbor2Neighbor

MAP-SNR = Mapping between Adjacent Pixels and eventually work for Speckle Noise Reduction

SNR = Signal-to-Noise Ratio

CNR = Contrast-to-Noise Ratio

ENL = Equivalent Number of Looks

What We Gained and Future Work

Developed a Novel speckle-free OCT imaging strategy

Demonstrated the Power of Teacher-Guided GAN in OCT Denoising

Utilization of this method on other imaging modalities

References

Ni G, Wu R, Zheng F, Li M, Huang S, Ge X, Liu L, Liu Y. Toward Ground-Truth Optical Coherence Tomography via Three-Dimensional Unsupervised Deep Learning Processing and Data. IEEE Trans Med Imaging. 2024 Jun;43(6):2395-2407. doi: 10.1109/TMI.2024.3363416. Epub 2024 Jun 3. PMID: 38324426.

Ma, Yuhui et al. “Speckle noise reduction in optical coherence tomography images based on edge-sensitive cGAN.” Biomedical optics express vol. 9,11 5129-5146. 2 Oct. 2018, doi:10.1364/BOE.9.005129

Liba, O., Lew, M., SoRelle, E. et al. Speckle-modulating optical coherence tomography in living mice and humans. Nat Commun 8, 15845 (2017). <https://doi.org/10.1038/ncomms15845>

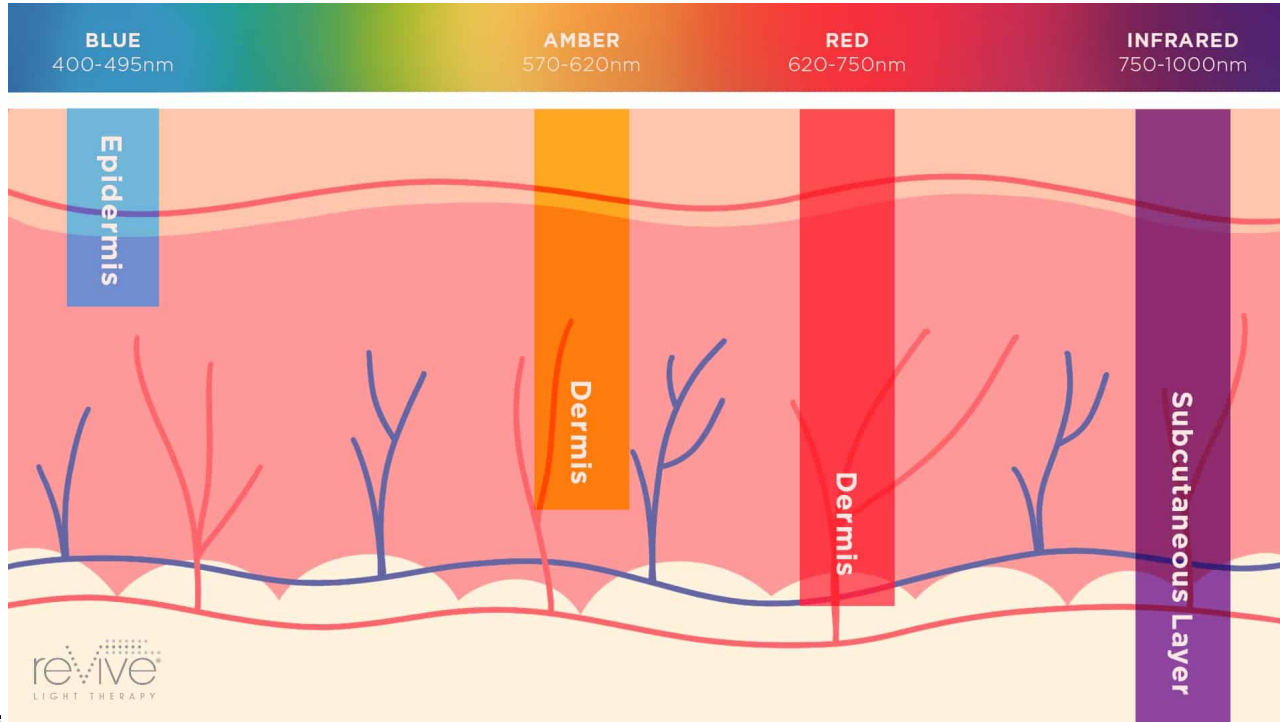
Hu, Y., Ren, J., Yang, J. et al. Noise reduction by adaptive-SIN filtering for retinal OCT images. Sci Rep 11, 19498 (2021). <https://doi.org/10.1038/s41598-021-98832-w>

Thank You For Listening!

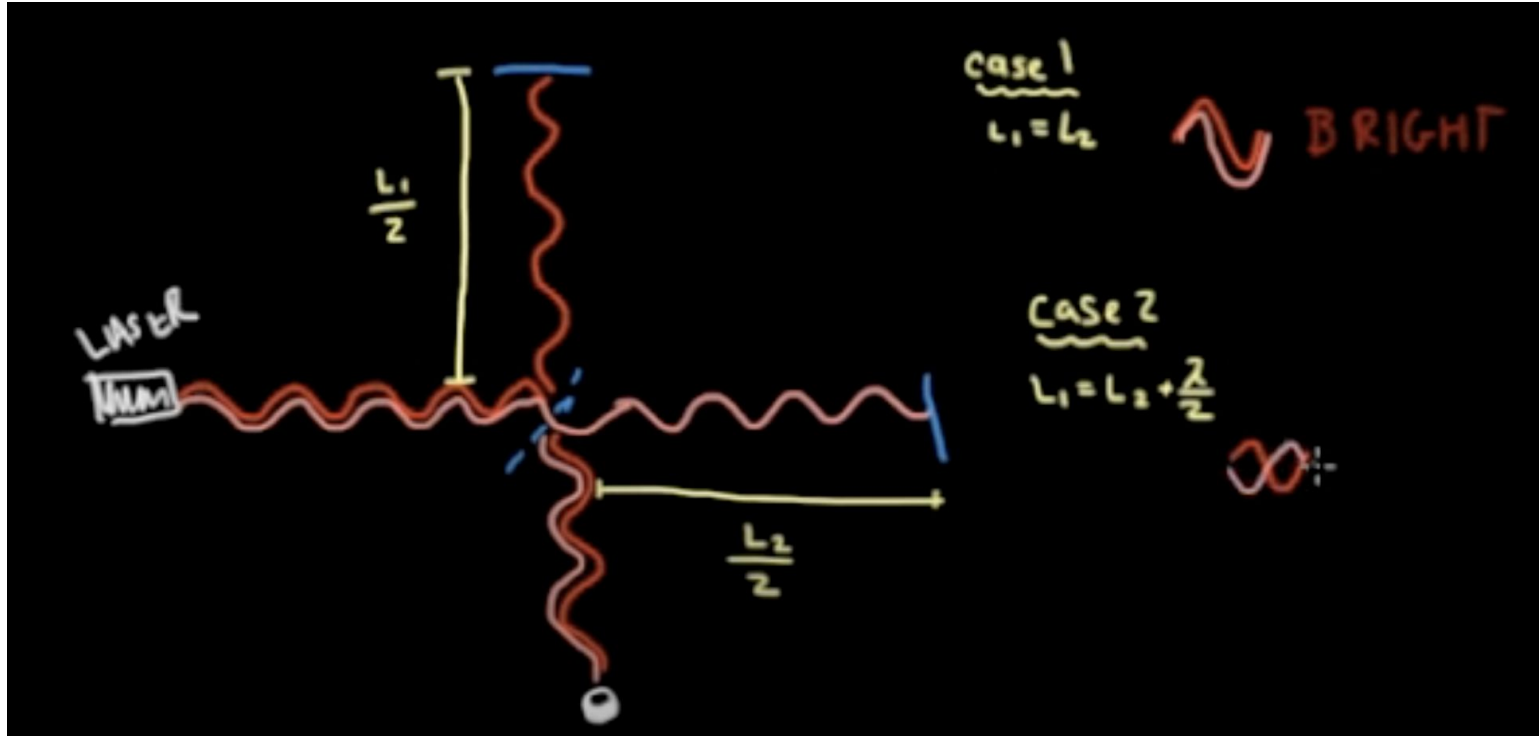
Appendix

Why BBS...

Broadband Source Used Because Different Wavelength Penetrate Deeper



The Interferometer in Question



Datasets

Dataset	OCT Setup	Axial displacement	Lateral displacement	Data size	Subject Application	
OCT-R1	BM-400K BMizar	~3.8 mm	10 mm	512×512×512	25	T (Training) &
				1948×1536×1280	16	V (Validating)
OCT-N1	Customized OCT	~1.68 mm	5.10 mm	800×800×800	46	T & V
OCT-R2	Spectralis OCT	~4.1 mm	11.8 mm	512×512×400	5	Testing
OCT-N2	Customized SM-OCT	~1.68 mm	5.10 mm	800×420×100	4	Testing

R = Retinal

N = Non-Retinal

Loss Functions - Extra Help

$$L_{MSE} = \frac{1}{HW} \sum_{i,j=1}^{H,W} \|x_{i,j} - y_{i,j}\|^2$$

Loss Functions - Extra Help

$$L_{Vgg} = \frac{1}{HW} \sum_{i,j=1}^{H,W} \|VGG_{16}(x)_{i,j} - VGG_{16}(y)_{i,j}\|^2$$

Loss Function for Generator Complete

$$L_G = \alpha L_{adv} + \beta L_{MSE} + \gamma L_{Vgg}$$

$$\alpha = 0.001 \quad \beta = 1 \quad \gamma = 1$$

Found Optimal Balance with Experimentation

tGT-OCT - Quantitative Comparisons

Table 2. Quantitative results of the proposed *t*GT-OCT compared with SM-OCT

		PSNR	SSIM	SNR	CNR	ENL
Scotch Tape	Noisy	14.451	0.163	15.546	1.684	0.360
	<i>t</i> GT-OCT	29.286	0.783	39.420	6.142	118.076
Pork Meat	Noisy	13.510	0.103	10.847	-0.305	0.320
	<i>t</i> GT-OCT	28.078	0.725	40.151	3.094	574.779
Fish Meat	Noisy	13.239	0.102	5.114	-3.098	0.402
	<i>t</i> GT-OCT	28.788	0.744	30.917	0.603	139.853

PSNR = Peak Signal-to-Noise Ratio

SSIM = Structural Similarity Index Measure

SNR = Signal-to-Noise Ratio

CNR = Contrast-to-Noise Ratio

ENL = Equivalent Number of Looks